

Paraneoplastic Syndromes from Head to Toe: Pathophysiology, Imaging Features, and Workup

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Patients often have symptoms due to the mass effect of a neoplasm on surrounding tissues or the development of distant metastases. However, some patients may present with clinical symptoms that are not attributable to direct tumor invasion. In particular, certain tumors may release substances such as hormones or cytokines or trigger an immune cross-reactivity between malignant and normal body cells, resulting in characteristic clinical features that are broadly referred to as paraneoplastic syndromes (PNSs). Recent advances in medicine have improved the understanding of the pathogenesis of PNSs and enhanced their diagnosis and treatment. It is estimated that 8% of patients with cancer develop a PNS. Diverse organ systems may be involved, most notably the neurologic, musculoskeletal, endocrinologic, dermatologic, gastrointestinal, and cardiovascular systems. Knowledge of various PNSs is necessary, as these syndromes may precede tumor development, complicate the patient's clinical presentation, indicate tumor prognosis, or be mistaken for metastatic spread. Radiologists should be familiar with the clinical presentations of common PNSs and the selection of appropriate imaging examinations. Many of these PNSs have imaging features that can assist with arriving at the correct diagnosis. Therefore, the key radiographic findings associated with these PNSs and the diagnostic pitfalls that can be encountered during imaging are important, as their detection can facilitate early identification of the underlying tumor, reveal early recurrence, and enable monitoring of the patient's response to therapy.

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Supplemental Material



Quiz questions for this article are available in the [supplemental material](#).

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Abbreviations: CECT = contrast-enhanced CT, FDG = fluorodeoxyglucose, HHM = humoral hypercalcemia of malignancy, HOA = hypertrophic osteoarthropathy, LE = limbic encephalitis, NBTE = nonbacterial thrombotic endocarditis, PCD = paraneoplastic cerebellar degeneration, PFPA = palmar fasciitis and polyarthritis, PNS = paraneoplastic syndrome, SCLC = small cell lung cancer, SIADH = syndrome of inappropriate antidiuretic hormone secretion

TEACHING POINTS

- In some patients, PNSs are the initial sign of an underlying tumor, prompting cancer screening and further workup. PNSs may also serve as the first sign of disease recurrence or progression. Thus, they can be used to monitor the evolution of the underlying tumor.
- It is estimated that 50% of patients older than 50 years with severe subacute ataxia have an underlying tumor.
- Patients with paraneoplastic myelitis frequently have an underlying SCLC or breast cancer, and the most common associated antibodies are anti-Hu, anti-CRMP5, and anti-Ri.
- Thymoma and thymic hyperplasia are both associated with myasthenia gravis; even patients with hyperplasia may undergo thymectomy.
- Osteomalacia is uncommon before the age of 40 years. Adults usually present with pathologic fractures and muscle weakness, while children present with gait abnormality, skeletal changes, and stunted growth.

Introduction

Paraneoplastic syndromes (PNSs) are a constellation of disease manifestations that are caused by an underlying neoplasm. However, they are not directly related to invasion by a primary or metastatic tumor (1). Immunologic or biochemical tumor-derived substances are often identified as a cause, although in many cases, the true causative factor may remain unknown. PNSs are thought to affect up to 8% of patients with cancer (2), and this prevalence will likely increase as cancer survival durations become longer (1).

In many malignant neoplasms, PNSs may manifest before a cancer diagnosis. Therefore, the timely recognition of a PNS may allow early detection and treatment of a previously unsuspected tumor, leading to improved clinical outcomes (1). In some patients, PNSs are the initial sign of an underlying tumor, prompting cancer screening and further workup (1). PNSs may also serve as the first sign of disease recurrence or progression. Thus, they can be used to monitor the evolution of the underlying tumor (2). PNSs should be included in the differential diagnosis in patients with constitutional symptoms (eg, weight loss), advanced age, subacute presentation, and progressive deterioration that is not explained by other causes or is not responsive to treatment (3,4), particularly if associated with other neuropathies (eg, somatic sensory neuropathy) (5), with evidence of known cancer risk factors (eg, smoking), or in the presence of supportive laboratory findings (eg, tumor markers and antibodies commonly associated with tumors) (6).

PNSs may involve diverse organ systems, including but not limited to the neurologic, musculoskeletal, dermatologic, endocrinologic, gastrointestinal, and cardiovascular systems.

Radiologists should be aware of the commonly encountered PNSs, familiar with their imaging features, and experienced in selecting the appropriate imaging examinations for patients with PNS (7). We discuss the imaging features of PNSs, associated causative malignant neoplasms (Table 1), and common diagnostic pearls and pitfalls that can help the radiologist to suggest a PNS diagnosis. We also review the European Federation of Neurological Societies' imaging recommendations for patients with PNS and review the imaging strategy for investigation in patients with unknown primary tumors (8).

Neurologic PNSs

In contrast to some other PNSs (eg, endocrinologic), neurologic PNSs develop before cancer detection in 80% of cases (1,9). Tumor-directed antibodies (ie, onconeural antibodies) are produced in response to a growing cancer and can be classified into two groups: those with a strong cancer association, such as anti-Hu (antineuronal nuclear antibody [ANNA] type 1), anti-CV2 (CRMP5), and anti-amphiphysin, and those with partial association to a specific cancer, such as anti-metabotropic glutamate receptor (mGluR) 1 and ANNA type 3 (ANNA-3) (1). Onconeural antibodies and their common associated PNSs are summarized in Table 2.

Detection of onconeural antibodies in cases of a possible PNS strongly indicates an underlying malignant neoplasm and helps in diagnosing a neurologic condition as a true PNS. However, the absence of antibodies in the presence of a neurologic syndrome should not be a reason to stop searching for an underlying tumor, as some antibodies might not be detected. If a tumor is identified but is not known to be associated with a PNS or onconeural antibodies, a search for an additional tumor should be performed (4).

Screening imaging examinations are usually indicated when a PNS is detected without an identifiable tumor (10). To detect an underlying tumor, CT of the chest, abdomen, and pelvis should be performed, followed by fluorodeoxyglucose (FDG) PET or PET/CT if the initial CT examination yields negative results. Neurologic PNSs can affect the central and peripheral nervous systems, as well as neuromuscular junctions, as described in the following sections.

PNSs of the Central Nervous System

Encephalitis.—Limbic encephalitis (LE) is a PNS that usually manifests with seizures, short-term memory loss, confusion, and psychologic dysfunctions that rapidly progress over 3 months (4). The most common underlying tumor is small cell lung cancer (SCLC), which is found in an average of 55% of patients with LE and is commonly associated with anti-Hu antibodies (4). LE has also been reported to develop in up to 13% of patients with testicular tumors with the presence of anti-Ta antibodies and 5% of patients with breast cancer (7). Anti-N-methyl-D-aspartate (NMDA)-receptor antibodies are found in LE in young women between 20 and 50 years of age and are strongly associated with tumors (59%), particularly with mature and immature ovarian teratomas (Fig E1) (11,12). On MR images, the earliest visible changes of LE appear as hyperintense signal with fluid-attenuated inversion-recovery and

Table 1: PNSs and Their Tumor Associations

PNS	Primary Tumor
Neurologic	
LE	SCLC, testicular germ cell, breast cancer, Hodgkin lymphoma
Anti-NMDA receptor encephalitis	Ovarian teratoma, SCLC, Hodgkin lymphoma
Cerebellar degeneration	SCLC, ovarian tumors, Hodgkin lymphoma, breast cancer
Autonomic neuropathy	SCLC, thymoma
Peripheral neuropathy	SCLC, breast cancer, ovarian cancer, soft-tissue sarcomas, Hodgkin lymphoma
Cancer-associated retinopathy	SCLC
Morvan syndrome, neuromyotonia	Thymoma
Opsoclonus-myoclonus syndrome	Neuroblastoma, SCLC, breast cancer, gynecologic tumors, renal tumors
Paraneoplastic myelitis	SCLC, breast cancer
Musculoskeletal	
Myopathy (Lambert-Eaton myasthenic syndrome)	SCLC, prostate cancer, cervical cancer, lymphoma
Myasthenia gravis	Thymoma
Hypertrophic osteoarthropathy	Pleural mesothelioma (most common), non-SCLC, nasopharyngeal cancer, esophageal cancer, breast cancer, thoracic lymphoma
Dermatomyositis	Breast cancer, lung cancer, ovarian cancer, prostate cancer
Osteogenic osteomalacia	Phosphaturic mesenchymal tumors, angiofibroma, osteosarcoma
Palmar fasciitis and polyarthritis	Breast cancer, gynecologic malignant neoplasms, prostate cancer, gastric cancer, pancreatic cancer
Gastrointestinal	
Paraneoplastic gastrointestinal dysmotility	SCLC, ovarian carcinoma, retroperitoneal lymphoma
Stauffer syndrome	Renal cell carcinoma, prostate cancer, soft-tissue sarcoma, bladder cancer
Endocrine	
Cushing syndrome	SCLC, bronchial carcinoid
Syndrome of inappropriate antidiuretic hormone secretion	SCLC, head and neck cancer, breast cancer, prostate cancer
Hypercalcemia	Squamous cell carcinoma, osteosarcoma, SCLC, Hodgkin lymphoma, renal tumors, gynecologic tumors
Cardiovascular	
Eosinophilic myocarditis	Lung adenocarcinoma, T-cell lymphoma, renal cell carcinoma
Nonbacterial thrombotic endocarditis	Lung adenocarcinoma, gastric adenocarcinoma, pancreatic cancer
Cardiac amyloidosis	Clonal plasma cell or other B-cell dyscrasias
Venous thromboembolism	Pancreatic tumors, brain cancer, ovarian cancer, hepatocellular carcinoma
Trousseau syndrome	Mucin-secreting adenocarcinomas (most common) of pancreatic, pulmonary, gastric, prostatic, or colonic origin

Note.—LE = limbic encephalitis, NMDA = *N*-methyl-D-aspartate, SCLC = small cell lung cancer

T2-weighted sequences, typically involving the medial temporal lobes and limbic structures (Figs 1, 2). Brain stem involvement has also been described. Over time, LE may cause mesial temporal atrophy (13).

Although LE can result from a nonparaneoplastic syndrome, the strong association with cancer warrants investigation for malignancy (4,7). In the setting of LE, the European Federation of Neurological Societies recommends contrast-enhanced CT (CECT) of the chest to detect SCLC, particularly if LE manifests in patients older than 40 years of age or in those with detectable anti-Hu antibodies (7,14). If cancer is not detected at chest CT, or in patients with LE who are younger than 40 years of age or who have negative results

for anti-Hu antibodies, then the imaging evaluation should be broadened to identify other malignancies. FDG PET is used for evaluation of a potential occult tumor not seen at CT (15).

The differential diagnosis for a medial temporal lobe T2-weighted fluid-attenuated inversion-recovery hyperintensity includes herpes simplex virus encephalitis, primary neoplasms of the central nervous system, and postictal changes. PNS and herpetic encephalitis can have a similar overall imaging appearance. Several findings can suggest herpes simplex virus, such as basal ganglia sparing and propensity for hemorrhage, as well as a differing clinical presentation. Herpes simplex virus tends to be more acute, and LE tends to manifest with psychiatric symptoms

Table 2: Summary of Onconeural Antibodies and their Associated PNSs and Underlying Tumors

Onconeural Antibodies	Neoplastic Association	Neurologic PNSs	Common Underlying Tumors
Anti-CV2/CRMP5	Strong	Encephalomyelitis, peripheral neuropathy	SCLC, thymoma
Anti-Hu	Strong	LE, encephalomyelitis, peripheral neuropathy	SCLC, non-SCLC, some neuroendocrine tumors
Anti-Ma2 and anti-Ma	Strong	Limbic and brain stem encephalitis	Testicular cancer, non-SCLC
Anti-mGluR5	Less strong	Encephalitis	Hodgkin lymphoma
Anti-NMDA receptor	Less strong	Anti-NMDA receptor encephalitis	Ovarian and extraovarian teratoma
Anti-P/Q voltage-gated calcium channel	Less strong	Lambert-Eaton myasthenic syndrome, cerebellar degeneration	SCLC
Anti-PCA2	Less strong	Peripheral neuropathy, cerebellar degeneration	SCLC, non-SCLC, breast cancer
Anti-Ri	Strong	Cerebellar degeneration, opsoclonus-myoclonus syndrome	Breast cancer, SCLC, non-SCLC
Anti-SOX1	Strong	Lambert-Eaton myasthenic syndrome, cerebellar degeneration	SCLC
Anti-Tr	Less strong	Cerebellar degeneration	Hodgkin lymphoma
Anti-Yo	Strong	Cerebellar degeneration	Ovary and breast cancer
Antiamphiphysin	Strong	Peripheral neuropathy, encephalomyelitis	SCLC, breast cancer

Note.—NMDA = *N*-methyl-D-aspartate, PCA2 = Purkinje cell antibodies, SOX-1 = SRY-box transcription factor 1.

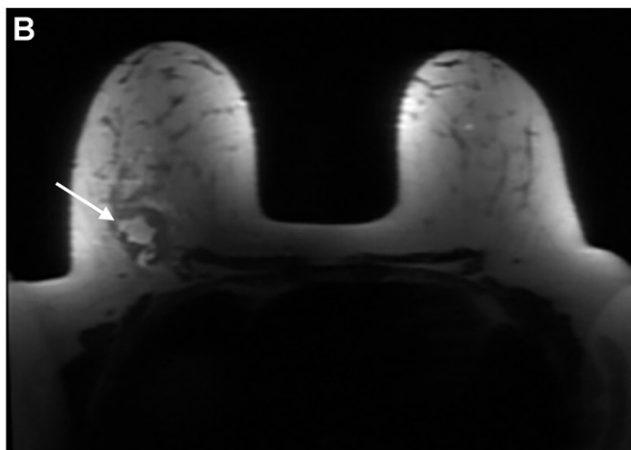
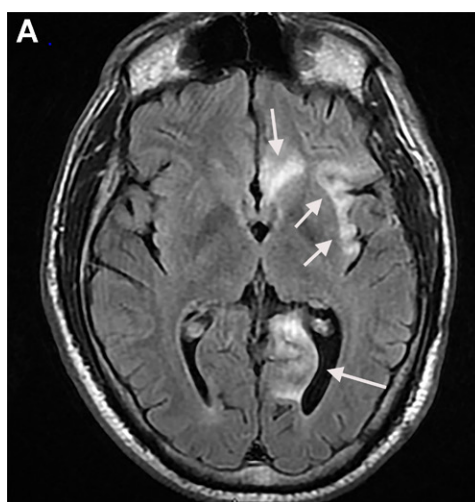


Figure 1. Autoimmune LE in a 53-year-old woman with newly diagnosed breast cancer. (A) Axial fluid-attenuated inversion-recovery MR image of the brain shows hyperintense signal in the left medial temporal lobe and temporal pole, posterior cingulate gyrus, insula, and orbito-frontal region (arrows). (B) Axial T2-weighted MR image of the breasts shows a mass in the right breast with an irregular rim and central necrosis (arrow).

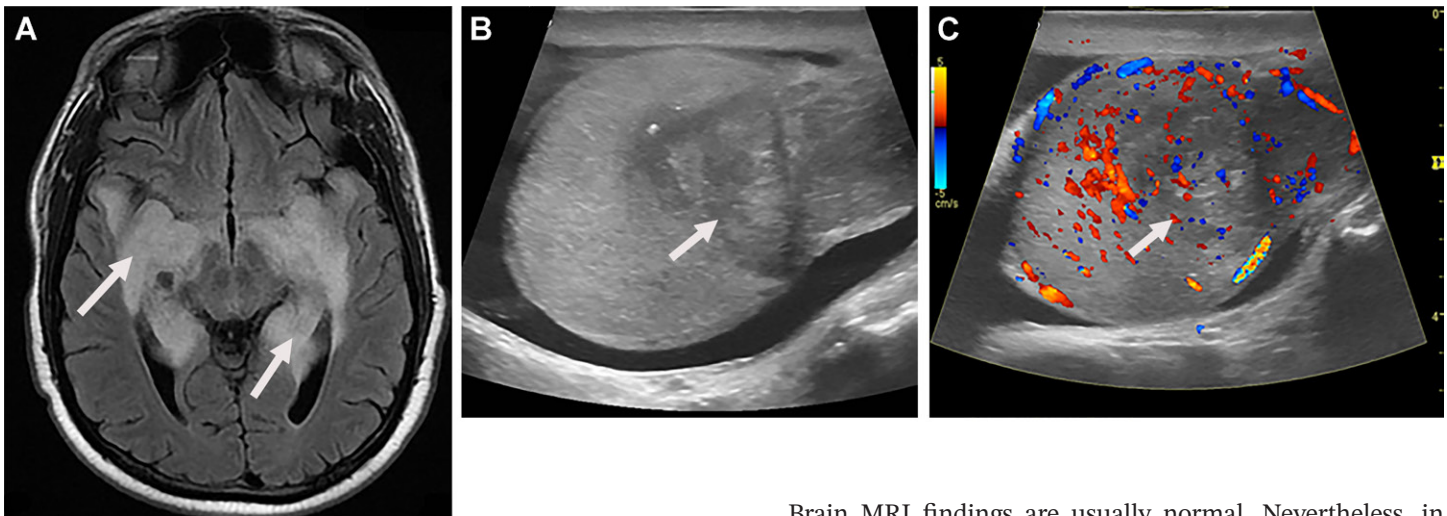
(13,16). High-grade glioma demonstrates a fairly characteristic imaging appearance with central necrosis. Radiologic differentiation between early stages of glioma and LE is difficult. However, MR spectroscopy can help in early identification by the increased choline-to-*N*-acetyl-aspartate ratio in neoplastic cases (17). A close MRI follow-up is recommended to avoid diagnostic pitfalls (17). While LE imaging findings are indistinguishable from postictal findings, an acute history of a complex seizure usually helps to reach the diagnosis (13).

Cerebellar Degeneration.—Paraneoplastic cerebellar degeneration (PCD) is one of the most devastating examples of a re-

mote neurologic complication of a tumor (18). Most patients develop a rapidly progressive course limiting daily living activities within 3 months (4). Gait ataxia is usually the initial symptom, followed by truncal and limb involvement. PCD may also include ataxic dysarthria or diplopia (18). Neurologic impairments are bilateral and often show asymmetry (19).

Early in the disease course, brain MRI findings are often unremarkable. However, cerebellar edema has been reported in some cases. In the late stages, cerebellar atrophy corresponding to permanent neural loss can be seen (18,20). Signal intensity abnormalities in the brain stem may be noted when PCD coexists with another paraneoplastic disorder (eg, brain stem encephalitis or LE) (18).

Figure 2. LE in a 27-year-old man with an underlying testicular mixed germ cell tumor. (A) Axial T2-weighted MR image of the brain shows hyperintense signal in the mesial temporal lobes with mild regional mass effect (arrows). (B, C) Gray-scale US image (B) of the right testicle shows a large predominantly solid heterogeneously hypoechoic mass (arrow), with increased vascularity on the color Doppler US image (C).



It is estimated that 50% of patients older than 50 years with severe subacute ataxia have an underlying tumor (18). The malignant neoplasms most commonly associated with PCD are SCLC (anti-Hu antibody), gynecologic tumors (anti-Yo antibody), breast cancer (anti-Yo antibody), and Hodgkin disease (anti-Tr antibody) (Figs 3, 4). When anti-Hu antibodies are detected, evaluation for lung cancer should start with chest CT. For a suspected gynecologic malignant neoplasm, the initial investigations are CT or US of the abdominopelvic cavity and measurement of tumor marker cancer antigen 125 (CA-125) levels (14). Similarly, breast cancer evaluation requires mammography supplemented by US or MRI (18). In onconeural antibody-positive patients with no tumor detected at conventional imaging, FDG PET is the next step (18,21).

PCD is a rare finding that mimics other causes of cerebellar degeneration, such as alcoholic cerebellar degeneration, infectious or postinfectious cerebellitis, antineoplastic medications (particularly phenytoin), and ataxia associated with anti-glutamic acid decarboxylase (22). Alcoholic ataxia shows vermis atrophy on brain images. In addition, speech and ocular motility are usually preserved, while symptoms of peripheral neuropathy and imaging findings of posterior column degeneration are commonly seen (23). Unlike PCD, ataxia associated with anti-glutamic acid decarboxylase has a slower progression and is associated with polyendocrine dysfunction (eg, pernicious anemia, thyroiditis, or insulin-dependent diabetes mellitus) (22).

Opsoclonus-Myoclonus Syndrome.—Opsoclonus-myoclonus syndrome is characterized by saccadic eye movement with action myoclonus of the trunk, limbs, and head (4). It is closely associated with neuroblastoma among children, while it is more commonly associated with SCLC or breast cancer in adults (14). Most cases have no onconeural antibodies. However, patients with underlying breast cancer and opsoclonus-myoclonus syndrome may have anti-Ri antibodies (7).

Brain MRI findings are usually normal. Nevertheless, in chronic cases, cerebellar atrophy with pontine tegmental abnormalities has been described (24). According to the European Federation of Neurological Societies Task Force guidelines, imaging with chest and abdominopelvic CT is recommended to identify pulmonary, gynecologic, and other visceral primary malignant neoplasms (7,14). Furthermore, mammography is indicated in women with opsoclonus-myoclonus syndrome. In children, whole-body scintigraphy with radiolabeled metaiodobenzylguanidine may be used to identify an occult neuroblastoma when conventional imaging results are negative (25).

Paraneoplastic Myelitis.—Paraneoplastic myelitis is a rare entity and may occur in isolation or with involvement of other areas (eg, encephalomyelitis) (26). Paraneoplastic myelitis symptoms include flaccid or spastic tetraparesis or paraparesis, bowel or bladder dysfunction, and sensory loss (27). Patients may develop cerebral symptoms (eg, headache, seizures) in the case of associated intracranial involvement (26).

Paraneoplastic myelitis demonstrates longitudinally extensive hyperintensity at T2-weighted imaging and enhancement at contrast-enhanced T1-weighted imaging, which makes MRI of the entire spine particularly helpful (Fig 5) (26). Lateral column involvement is a characteristic finding, best seen in the axial plane (26). Abnormal FDG uptake in the involved spinal cord area has also been described and could be mistaken for tumor metastasis (28). Imaging of the brain is usually indicated to identify associated encephalitis, which is commonly seen in patients with positive results for anti-CRMP5 antibodies (26).

Patients with paraneoplastic myelitis frequently have an underlying SCLC or breast cancer, and the most common associated antibodies are anti-Hu, anti-CRMP5, and anti-Ri (27). Cerebrospinal fluid analysis may confirm elevated protein levels, pleocytosis, and oligoclonal bands. Despite treatment, most patients have a poor outcome and require the use of a wheelchair.

Differential considerations for paraneoplastic myelitis are broad and include transverse myelitis, radiation myelitis, spinal dural arteriovenous fistula, spinal cord infarct, and demyelinating disease, such as neuromyelitis optica or multiple sclerosis. Evidence of an enlarged vessel with a vascular flow

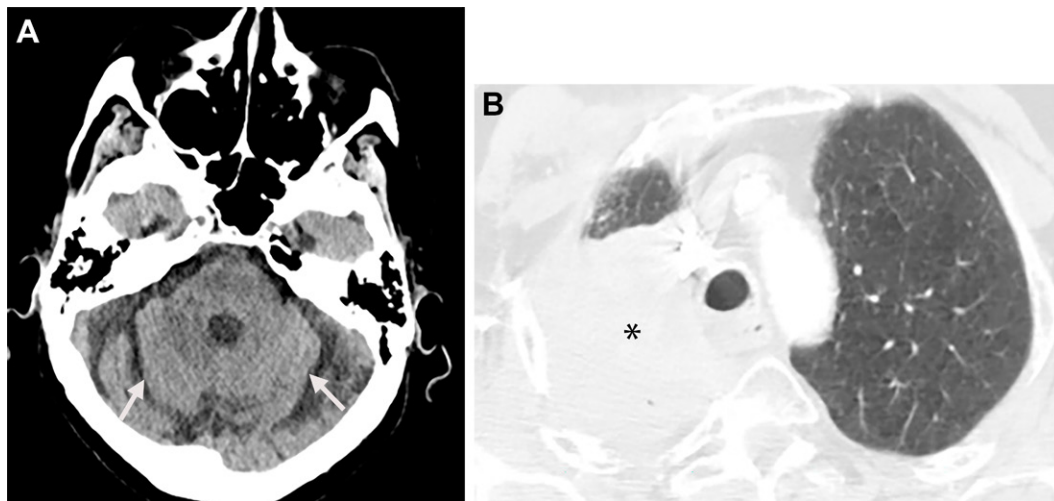


Figure 3. Cerebellar degeneration in a 66-year-old man with a history of lung cancer. (A) Axial nonenhanced CT image of the brain shows significant volume loss of the cerebellum (arrows). (B) Axial CECT image of the chest at the level of the aortic arch shows near-complete opacification of the right lung apex (*) indicative of postobstructive right upper lobe consolidation due to lung cancer.

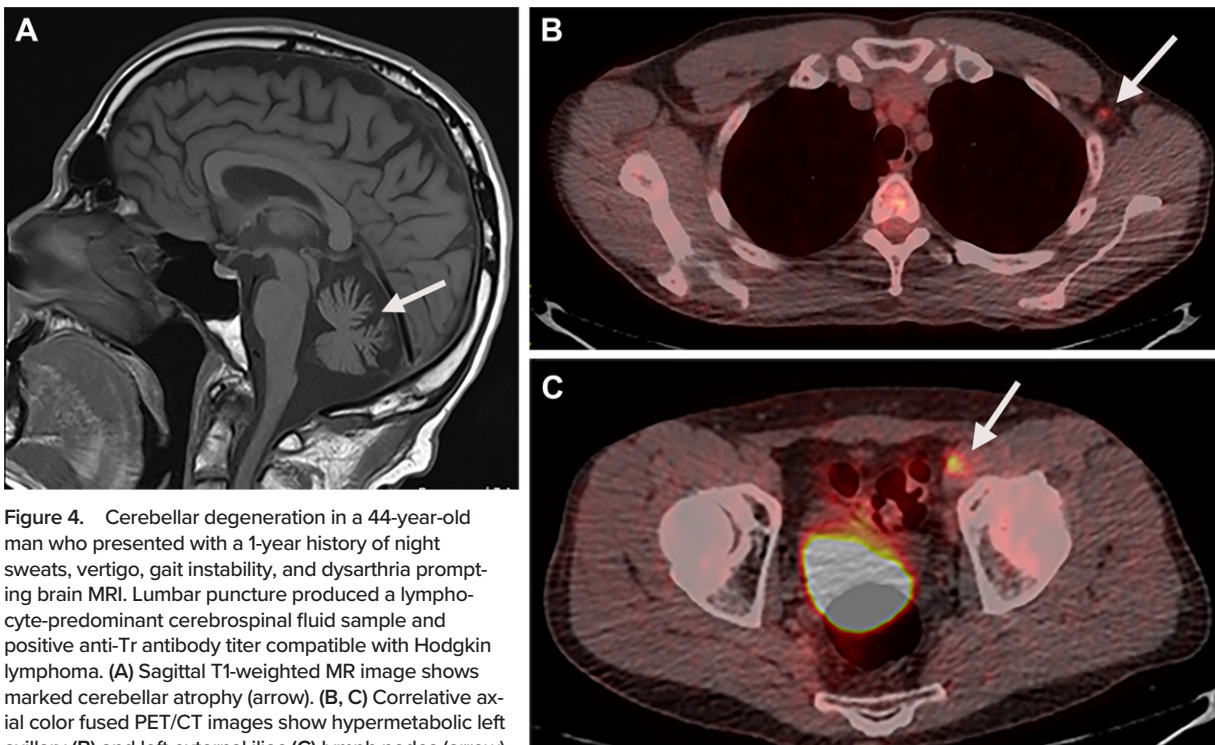


Figure 4. Cerebellar degeneration in a 44-year-old man who presented with a 1-year history of night sweats, vertigo, gait instability, and dysarthria prompting brain MRI. Lumbar puncture produced a lymphocyte-predominant cerebrospinal fluid sample and positive anti-Tr antibody titer compatible with Hodgkin lymphoma. (A) Sagittal T1-weighted MR image shows marked cerebellar atrophy (arrow). (B, C) Correlative axial color fused PET/CT images show hypermetabolic left axillary (B) and left external iliac (C) lymph nodes (arrow).

void posterior to the cord would suggest an arteriovenous fistula, while preferential anterior horn involvement that may show diffusion restriction is an indicator of spinal cord infarct. Although spinal cord involvement by multiple sclerosis shows hyperintense T2 signal, it would not generally have longitudinally extensive involvement (26).

PNSs of the Peripheral Nervous System

The peripheral nervous system is the initial site of involvement in one-third of patients diagnosed with PNS (29). Subacute sensory neuropathy is the most common pattern of paraneoplastic neuropathy and may occur along with motor

weakness or autonomic involvement (29). In other cases, only symptoms of autonomic neuropathy are noted (30). Patients with subacute sensory neuropathy present with either painful dysesthesia and mechanical hyperalgesia or sensory ataxia and loss of proprioceptive sensation.

Paraneoplastic cranial neuropathy is rare but has been described in several reports. Symptoms depend on the affected nerve, such as balance disorder and sensorineural hearing loss involving the cochlear nerve. Moreover, involvement of multiple cranial nerves has rarely been reported (31).

Subacute sensory neuropathy and autonomic neuropathy in PNS are most commonly associated with SCLC but may also

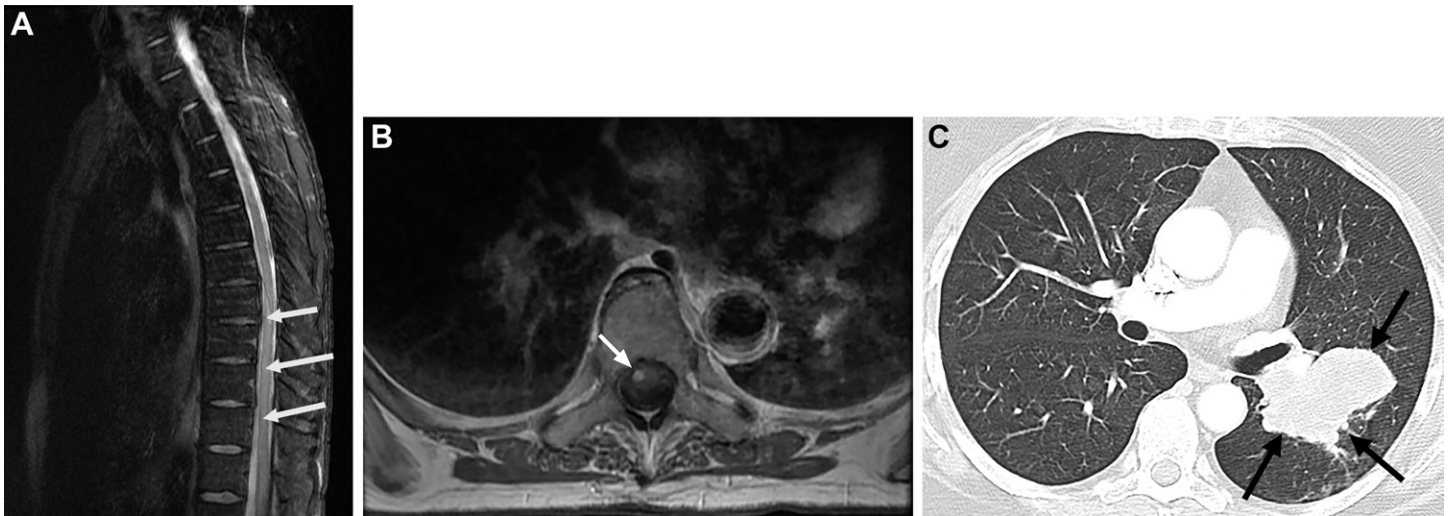


Figure 5. Paraneoplastic myelitis in a 49-year-old woman with bilateral lower limb weakness and gait abnormality and a known history of SCLC. (A) Sagittal T2-weighted fat-suppressed MR image of the thoracolumbar spine is a representative image of focally increased signal intensity in the dorsal cord that extends from the lower cervical spine to the upper thoracic spine and also from the mid to the lower thoracic spine (arrows). Myelitis was diagnosed on the basis of positive paraneoplastic panel results and no significant change over multiple examinations. (B) Axial postcontrast T1-weighted fat-suppressed MR image at the midthoracic level shows avid enhancement of the lateral column (arrow). (C) Axial CECT image of the chest just below the level of the carina shows a central lobular mass within the left lower lobe that extends into the left upper lobe (arrows), which was confirmed to be SCLC at pathologic examination. Based on a combination of clinical and investigative laboratory findings, an empirical diagnosis of paraneoplastic myelitis was assigned.

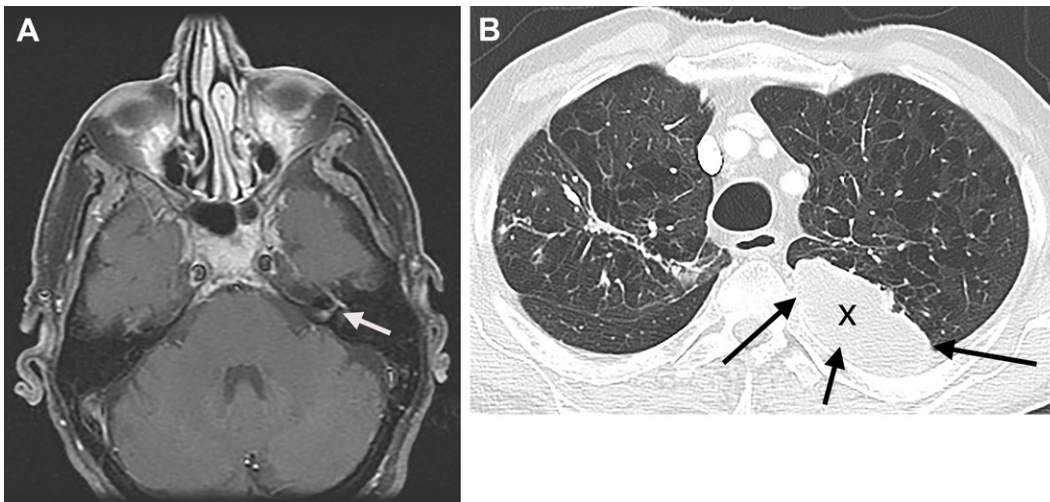


Figure 6. Cranial neuropathy in a 55-year-old man with a history of lung cancer. (A) Axial contrast-enhanced T1-weighted MR image shows abnormal enhancement and thickening of the left facial nerve (arrow) from the distal internal auditory canal to the geniculate ganglion. (B) Axial CECT image of the chest shows a large left upper lobe subpleural mass (arrows) with chest wall invasion and central necrosis (x).

occur with ovarian and breast cancers, Hodgkin lymphoma, and sarcoma (26), while paraneoplastic cranial neuropathy is more often associated with seminoma (32). Anti-Hu (7) and anti-CV2/CRMP5 are the most frequently detected antibodies in all paraneoplastic neuropathy cases (29).

At imaging, subacute sensory neuropathy demonstrates a variable enhancement pattern. At MRI, dorsal or ventral spinal nerve roots usually enhance on postcontrast T1-weighted images; this pattern does not necessarily correlate with patient symptoms (26). In cranial neuropathy, enhancement and/or enlargement of the affected nerves is noted (Fig 6).

The differential diagnosis of paraneoplastic neuropathy includes leptomeningeal carcinomatosis and neurosarcoidosis. However, these tend to have more nodular enhancement. Guillain-Barré syndrome could mimic paraneoplastic neuropathy and typically shows preferential ventral nerve root

enhancement. Chronic inflammatory demyelinating polyneuropathy could also be considered. However, it would have more prominent nerve enlargement and would be less likely to show any enhancement (26).

PNSs of the Neuromuscular Junction

Myasthenia Gravis.—Patients with myasthenia gravis report ocular and oropharyngeal symptoms that progress to proximal limb weakness and are commonly associated with thymoma. Chest CT is advised in patients with myasthenia gravis, especially those with positive findings for antiacetylcholine and ryanodine receptor autoantibodies. Thymoma and thymic hyperplasia are both associated with myasthenia gravis. Even patients with hyperplasia may undergo thymectomy (33).

Lambert-Eaton Myasthenic Syndrome.—Lambert-Eaton myasthenic syndrome (LEMS) is characterized by progressive

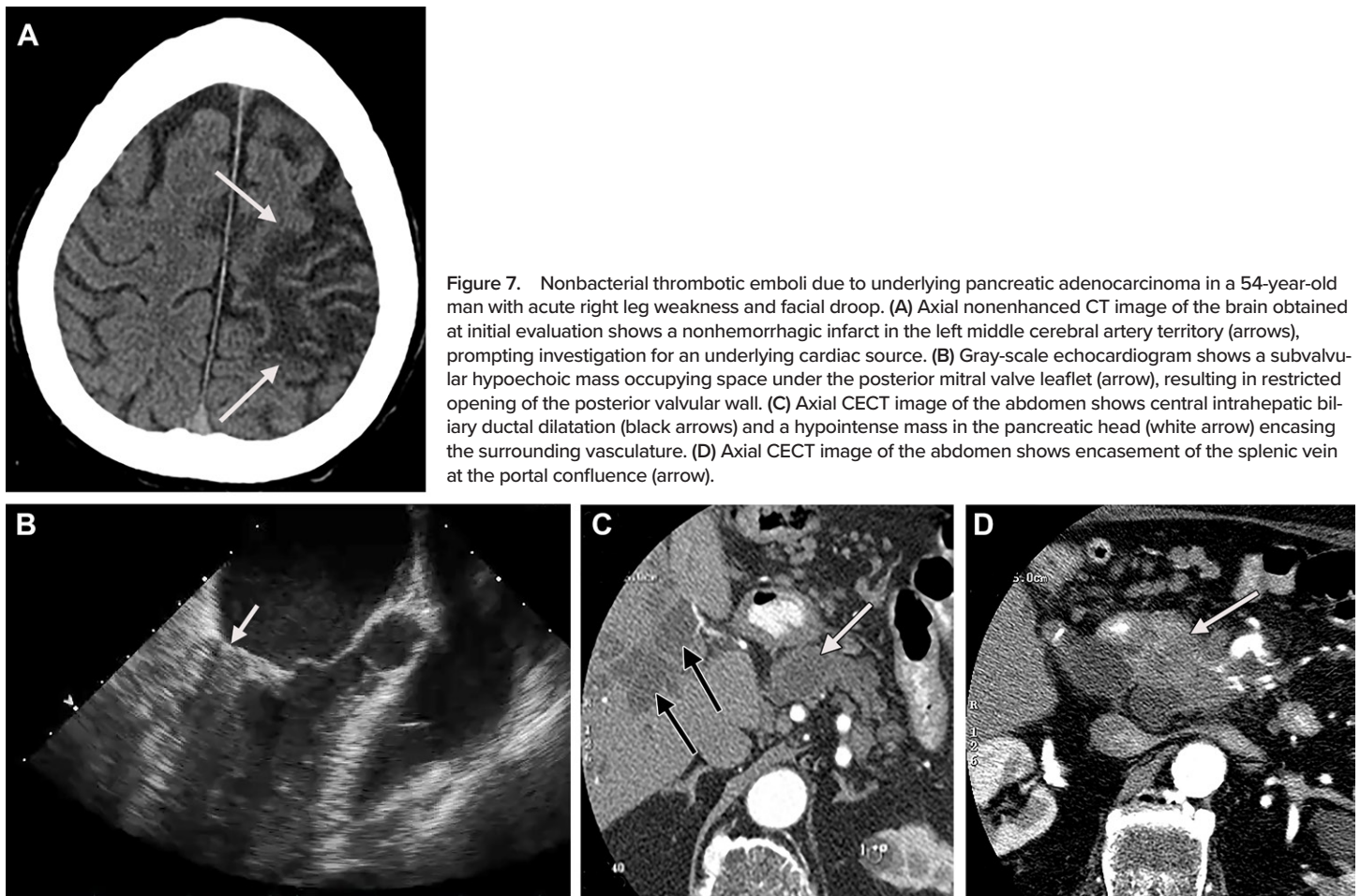


Figure 7. Nonbacterial thrombotic emboli due to underlying pancreatic adenocarcinoma in a 54-year-old man with acute right leg weakness and facial droop. (A) Axial nonenhanced CT image of the brain obtained at initial evaluation shows a nonhemorrhagic infarct in the left middle cerebral artery territory (arrows), prompting investigation for an underlying cardiac source. (B) Gray-scale echocardiogram shows a subvalvular hypoechoic mass occupying space under the posterior mitral valve leaflet (arrow), resulting in restricted opening of the posterior valvular wall. (C) Axial CECT image of the abdomen shows central intrahepatic biliary ductal dilatation (black arrows) and a hypointense mass in the pancreatic head (white arrow) encasing the surrounding vasculature. (D) Axial CECT image of the abdomen shows encasement of the splenic vein at the portal confluence (arrow).

proximal muscle weakness that starts in the lower limbs with autonomic involvement (eg, erectile dysfunction and dry mouth) (34). LEMS has significant association with SCLC and less commonly with breast, pancreatic, and ovarian carcinoma (7). All patients with LEMS should be examined with chest CT. If the CT findings are negative, PET/CT and bronchoscopy should be considered, particularly in patients who smoke and patients with positive test results for human leukocyte antigen B8 (HLA-B8) (7,14). Moreover, follow-up CT is recommended every 6 months for 4 years because of the high risk of cancer development (7).

Cardiovascular and Hematologic PNSs

Eosinophilic Myocarditis

Eosinophilic myocarditis is a rare type of myocardial inflammation characterized by eosinophilic infiltration of the cardiac muscles that may result in cardiogenic shock (35). It can occur from idiopathic hypereosinophilic syndrome or eosinophilic granulomatosis with polyangiitis, or it can be a PNS from a solid or hematologic malignancy. Patients present with worsening dyspnea, systolic hypotension, and, in some cases, jugular venous distention. Chest radiographs show cardiomegaly with bilateral pleural effusion, although it may be unremarkable in early cases with borderline systolic function. MR images show diffuse subendocardial late gadolinium enhancement, often with apical thrombus, and myocardial signal intensity increase with T2-weighted sequences (35,36). Contrarily, acute myocardial infarction demonstrates subendocardial late gadolinium enhancement based on the coronary artery disease territory, while cardiac amyloidosis demonstrates diffuse late gadolinium enhancement without T2 hyperintensity (36). Paraneoplastic disease with cardiac involvement has been reported with lung adenocarcinoma, T-cell lymphoma, and renal cell carcinoma (35).

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Nonbacterial Thrombotic Endocarditis

Nonbacterial thrombotic endocarditis (NBTE), formerly known as marantic endocarditis, is defined as deposition of thrombi consisting of fibrin strands and platelets on undamaged cardiac valves (37). This phenomenon has a well-known association with malignancy, autoimmune disorders, and some other systemic diseases. Clinical features result from the dislodgment of vegetative material, leading to systemic emboli circulating to the spleen, brain, and coronary vessels. NBTE should be strongly suspected in any patient who has stroke symptoms and has a known or suspected malignant neoplasm (37). Adenocarcinomas of the lung, stomach, and pancreas are the most common tumor types associated with NBTE (Fig 7, Movie 1).

Other pathologic abnormalities that may mimic NBTE include but are not limited to infective endocarditis, cardiac

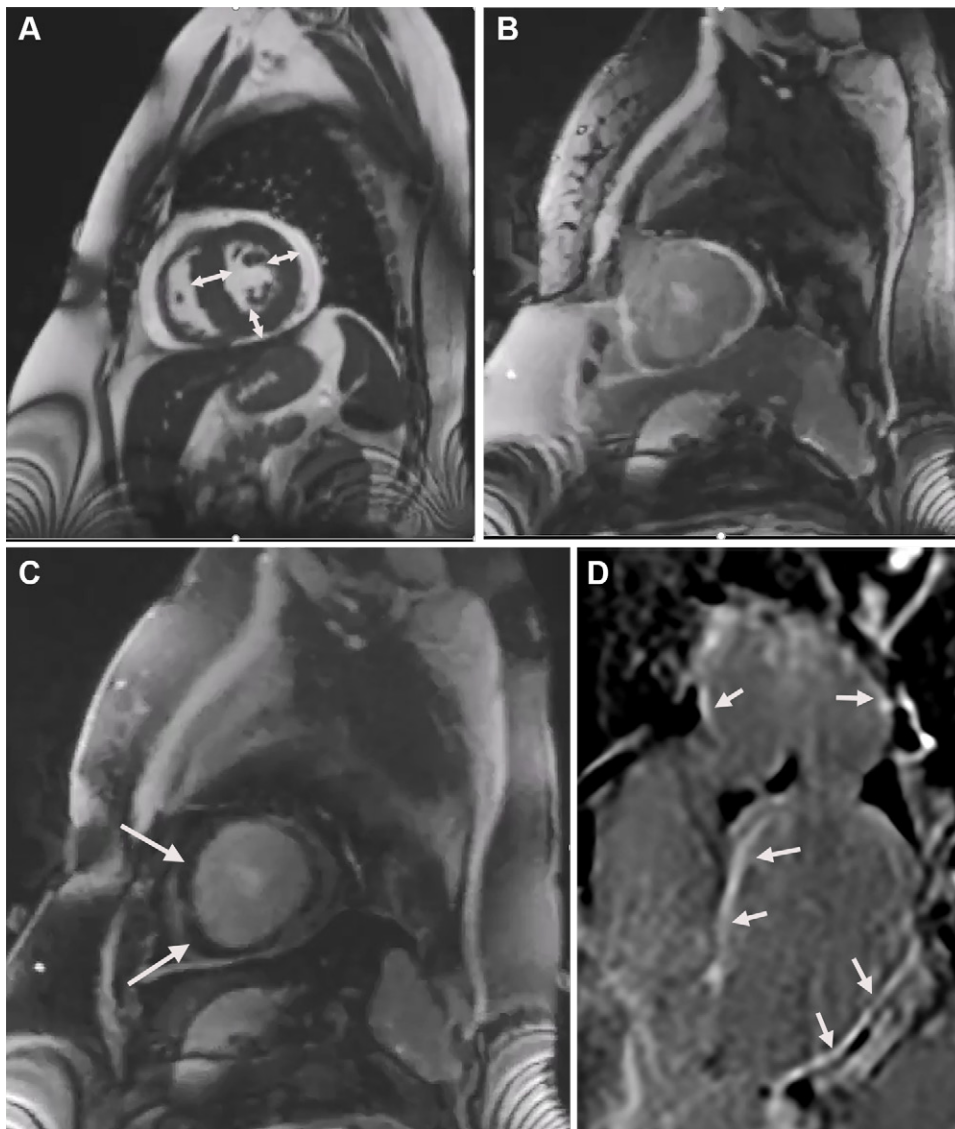


Figure 8. Cardiac amyloidosis in two patients. (A–C) Cardiac amyloidosis in a 61-year-old woman who presented with decreased energy for 1 year and nonoliguric renal failure. Short-axis stack MR time-to-inversion scout image (A) of the heart shows diffusely thickened left ventricular myocardium (double-headed arrows). Short-axis MR time-to-inversion scout images (B, C) at delayed time stamps show enhancement of the blood (B) and nulling of the myocardium earlier (C) than nulling of the blood pool (arrows in C), which is the reversal of the normal pattern. (D) Four-chamber delayed enhancement MR image in an 89-year-old woman with amyloidosis shows circumferential subendocardial abnormal delayed gadolinium enhancement of the left ventricle, with similar subendocardial enhancement in the atria (arrows).

myxoma, and aortic thrombus. However, NBTE is more likely if there is evidence of weight loss, resistance to an antibiotic treatment, and negative findings with the Duke criteria for infective endocarditis (38). Moreover, the patterns of embolic strokes seen at diffusion-weighted MRI of the brain may be able to be used to distinguish between NBTE and infective endocarditis as the underlying cause. In NBTE, embolic strokes typically exhibit multiple disseminated small to large infarcts, while infective endocarditis-related emboli show all stroke patterns (39). Therefore, a single embolic lesion or multiple lesions in the same arterial territory are more often seen in infective endocarditis (37,39). This could be explained by the little cellular component in NBTE vegetation leading to easier fragmentation and embolization (37).

Cardiac Amyloidosis

Amyloidosis is caused by the systemic deposition of misfolded proteins, leading to significant morbidity and mortality. The most common types are amyloid light chain and

transthyretin amyloidosis. However, only amyloid light chain amyloidosis occurs secondarily to a neoplastic process such as clonal plasma cell or other B-cell dyscrasias (40). Clinical manifestations include dyspnea, syncope, and diastolic heart failure. MR images show atrial enlargement, biventricular hypertrophy, and pericardial and pleural effusions. Other specific features are reversed nulling times of the myocardium and blood pool, leading to subendocardial hyperenhancement and hypoattenuation of the blood pool (Fig 8, Movie 2) (40).

Radionuclide scintigraphy using technetium 99m diphosphono-propanodicarboxylic acid can be performed to identify transthyretin amyloidosis and help differentiate it from amyloid light chain amyloidosis. Patients with transthyretin amyloidosis will have grade 2 or 3 tracer uptake in the heart at bone scintigraphy, as well as absence of detectable monoclonal gammopathy in the blood and urine, as seen in amyloid light chain amyloidosis (41). MRI could also play a role in amyloidosis type differentiation, as transthyretin amyloidosis has more left ventricular thickening, lower T1 signal

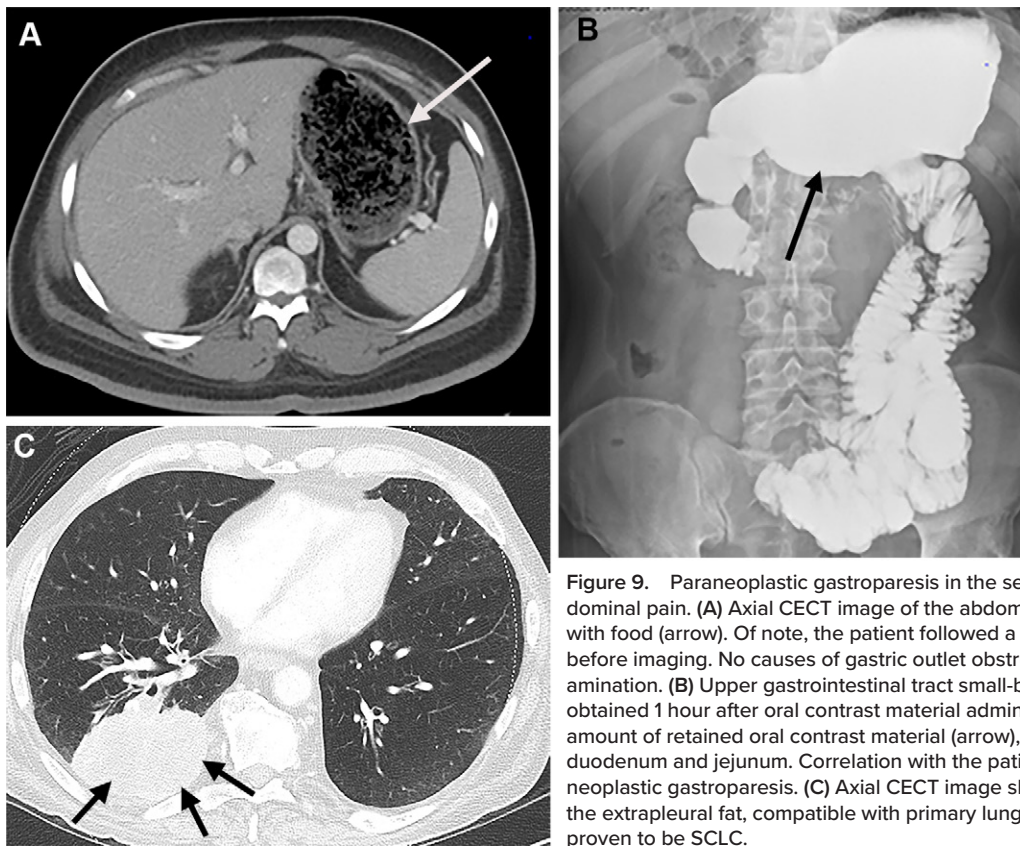


Figure 9. Paraneoplastic gastroparesis in the setting of SCLC in a 43-year-old man with abdominal pain. (A) Axial CECT image of the abdomen shows the stomach markedly distended with food (arrow). Of note, the patient followed a “nothing-per-mouth” (ie, nil per os) protocol before imaging. No causes of gastric outlet obstruction were identified at this imaging examination. (B) Upper gastrointestinal tract small-bowel follow-through fluoroscopic image obtained 1 hour after oral contrast material administration shows gastric distention with a large amount of retained oral contrast material (arrow), despite passage of contrast material into the duodenum and jejunum. Correlation with the patient’s history of cancer was indicative of paraneoplastic gastroparesis. (C) Axial CECT image shows a right lower lobe mass extending into the extrapleural fat, compatible with primary lung cancer (arrows), which was pathologically proven to be SCLC.

intensity, and more extensive late gadolinium enhancement than those of amyloid light chain amyloidosis (41).

Venous Thromboembolism

Venous thromboembolism in the absence of a conventional risk factor is a significant indicator of an occult tumor (42). It has notably been observed with tumors of the pancreas, brain, and ovary and with hepatocellular carcinoma. Physical examination, blood tests, and chest radiography have been shown to be helpful in identifying most cancers in patients with venous thromboembolism. Other extensive screening by using US and CT is not cost-effective and not justified except in the presence of symptoms concerning for an underlying malignancy (eg, change in bowel habits and weight loss) (7,42).

Trousseau Syndrome

Trousseau syndrome is a recurrent cancer-induced migratory venous thrombosis. This condition is characterized by a combination of migratory thrombophlebitis that causes painful erythematous rash and blood hypercoagulability that leads to development of venous and arterial thromboses. An underlying malignant neoplasm is diagnosed in 85%–95% of patients with Trousseau syndrome (43). Mucin-secreting adenocarcinomas are the most common malignant neoplasms associated with Trousseau syndrome, including those of the pancreas (50%), lung, stomach, prostate, and colon (44). Batsis et al (43) suggested the use of PET to identify occult tumors in patients with Trousseau syndrome because it has higher sensitivity and specificity than those of conventional imaging and could

cost less than extensive conventional investigations (such as multiple CT examinations, gastrointestinal endoscopy, and biopsy). Stroke in the setting of a three-territory involvement pattern at diffusion-weighted MRI (ie, three-territory sign) is highly specific for Trousseau syndrome and is six times more frequently observed in malignancy-associated stroke than in atrial fibrillation-associated stroke (45,46).

Gastrointestinal PNSs

One example of a gastrointestinal PNS is paraneoplastic gastrointestinal dysmotility, such as intestinal pseudo-obstruction, gastroparesis, and pseudoachalasia (5). It has been associated with several tumors, most commonly SCLC (anti-Hu) (Fig 9), ovarian carcinoma (anti-Yo), and retroperitoneal lymphoma (N-type calcium channel antibody) (5). In up to 80% of patients, paraneoplastic gastrointestinal dysmotility precedes the identification of malignancy (47). Patients with paraneoplastic gastrointestinal dysmotility often have other neuropathies, such as sensorimotor or autonomic neuropathy or encephalopathy, that suggest an underlying tumor (5). In addition, the occurrence of new-onset or prominent dysmotility symptoms should alert for an occult malignant neoplasm and indicate the need for further diagnostic tests (5). Initial investigations should include a gastric emptying study, esophageal manometry, autoimmune antibodies panel, and chest radiography to detect SCLC (5). In a patient at increased risk for lung mass with negative chest radiography and CT examinations, bronchoscopy and chest MRI are justified (5).

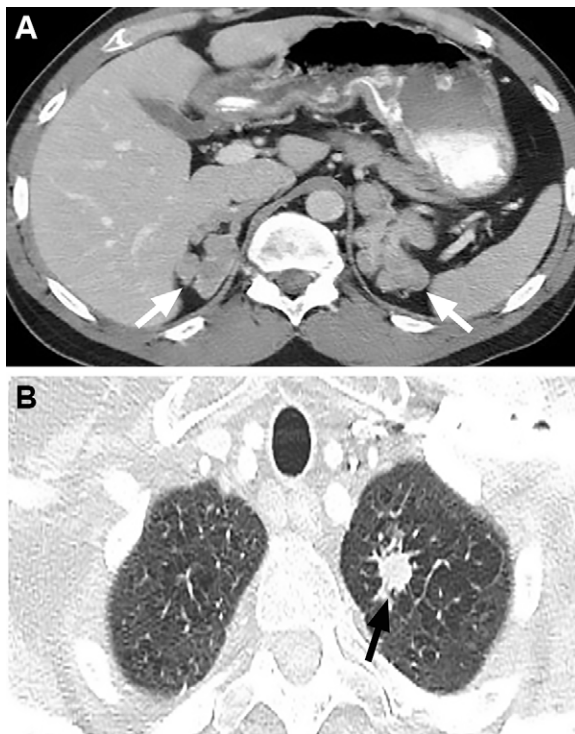


Figure 10. Cushing syndrome in a 62-year-old man with underlying lung cancer who presented with truncal obesity and facial fullness. (A) Axial CECT image of the upper abdomen shows bilateral diffuse adrenal enlargement (arrows) as a result of ectopic adrenocorticotrophic hormone production from SCLC. The diagnosis of adrenocorticotrophic hormone–dependent Cushing syndrome was made based on clinical, biochemical, and hormonal criteria. (B) Axial CECT image of the chest shows a left upper lobe spiculated solid mass (arrow) corresponding to biopsy-proven SCLC.

Endocrinologic PNSs

Endocrine PNSs arise from tumor production of peptides that lead to metabolic abnormalities (48). These include Cushing syndrome, inappropriate antidiuretic hormone secretion, and hypercalcemia.

Cushing Syndrome

Approximately 5%–10% of cases of Cushing syndrome, or hypercortisolism, are considered to be of paraneoplastic origin, with almost 50% of these cases secondary to neuroendocrine lung tumors (SCLC and bronchial carcinoids) (1,48). Cushing syndrome may also develop from neuroendocrine tumors of the thymus, adrenal glands, or gastrointestinal tract and rarely develop from those of the ovary or prostate (48). Patients with paraneoplastic Cushing syndrome often present with hypertension, muscle weakness, symptoms of hypokalemia, and generalized edema, while patients with nonparaneoplastic Cushing syndrome are more likely to report weight gain with centripetal fat distribution (49). Failure to respond to high-dose dexamethasone suppression confirms ectopic paraneoplastic Cushing syndrome rather than a nonparaneoplastic pituitary source (1,49).

Imaging examinations are required to locate the primary tumor and rule out a pituitary or adrenal lesion. Adrenal gland enlargement and primary tumor identification can

be noted at conventional imaging (eg, CT [Fig 10]) (50). Concordant increased uptake within the primary tumor at somatostatin receptor scintigraphy (eg, octreotide) permits the starting of treatment (51). In patients with a positive CT examination and negative octreotide imaging examination, FDG PET is the third advised examination (50). Gallium tetraazacyclododecane tetraacetic acid–octreotate (DOTATATE) imaging offers the highest sensitivity but is much less available than octreotide imaging. Therefore, these examinations are justified only for aggressive tumors and patients with discordant findings (50).

Syndrome of Inappropriate Antidiuretic Hormone Secretion

Syndrome of inappropriate antidiuretic hormone secretion (SIADH) is characterized by euvoletic hyponatremia and occurs in 1%–2% of all patients with malignant tumors (48). The symptoms of SIADH depend on the rapidity of onset and the degree of hyponatremia, which may result in weakness, headache, and memory difficulties. In severe cases, patients may develop seizures, coma, and respiratory collapse (52).

Approximately 10%–45% of patients with SCLC develop SIADH (Fig 11) (53). A wide variety of other tumors have also been found in conjunction with SIADH, including head and neck, breast, and prostate tumors (48). Benign conditions underlying SIADH include intrathoracic infection and central nervous system disorders.

In patients with biochemically confirmed unexplained SIADH, chest radiography and CT are essential for identifying both malignant and nonmalignant pulmonary causes. If chest imaging examinations yield negative results, depending on clinical presentation, neuroimaging may be indicated to identify a benign intracranial cause (7,54).

Hypercalcemia

Humoral hypercalcemia of malignancy (HHM) occurs in up to 10% of all patients with advanced tumors and generally signals a poor prognosis (1). Hypercalcemia manifestations include nausea, vomiting, confusion, polyuria, renal failure, and coma (55). Symptom severity is determined by the degree of hypercalcemia, rapidity of onset, and the patient's baseline renal and neurologic function (1). The most frequent tumor associated with HHM is squamous cell carcinoma. However, underlying osteosarcoma, SCLC, Hodgkin lymphoma, and renal and gynecologic malignant neoplasms have also been reported (51) (Fig 12). Before the consideration of an HHM diagnosis, benign causes such as sarcoidosis and hyperparathyroidism should first be excluded.

The primary tumor secretes parathyroid hormone–related protein in 80% of HHM cases (51). A multimodality imaging approach is recommended to evaluate HHM. Chest radiography can show primary lung cancer or granulomatous disorder (eg, sarcoidosis) (55). Skeletal scintigraphy and bone scans can be used to assess the extent of metastatic spread in the skeleton. Although skeletal scintigraphy is proven to be an effective modality in detecting bone metastases, FDG PET has been shown to be more accurate, especially in regard to hematologic malignancies (56). Other PET tracers are preferred for

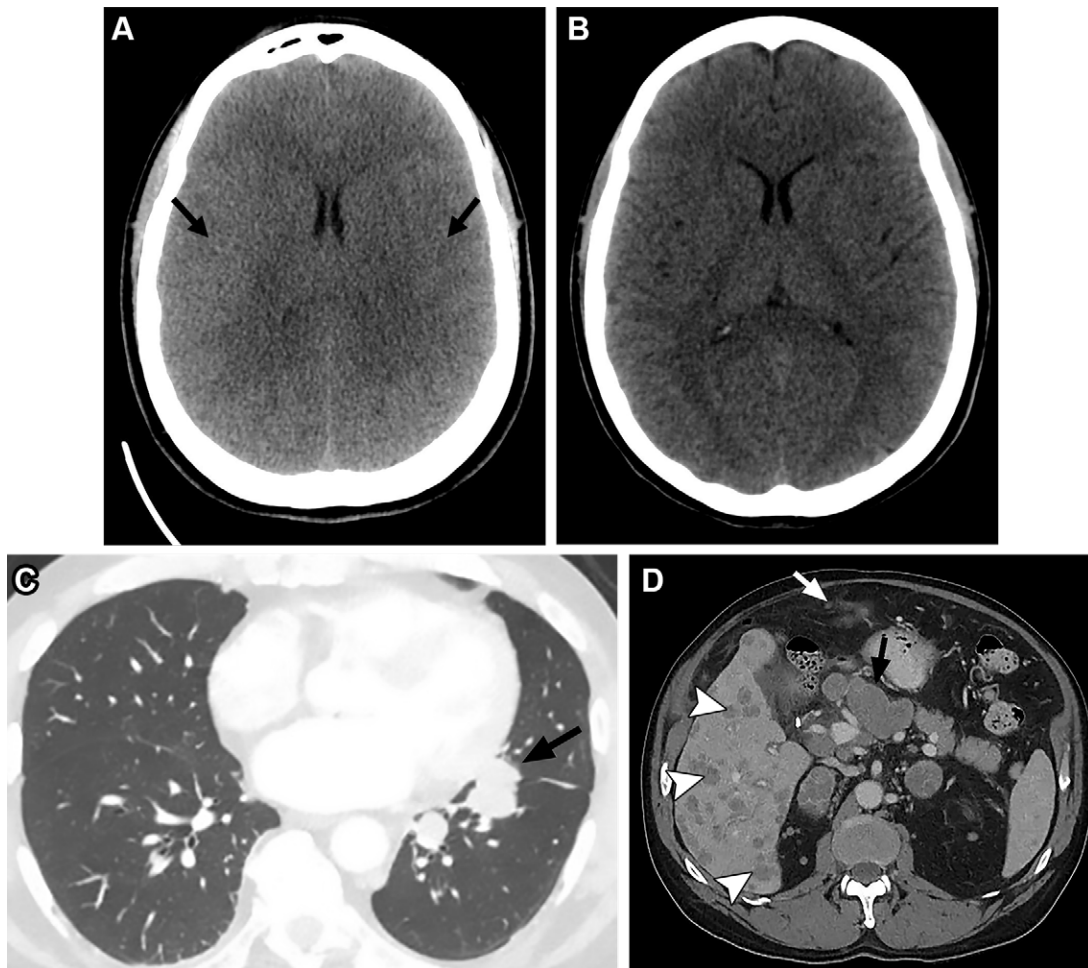


Figure 11. SIADH and hyponatremia in a 48-year-old woman who presented with seizures. (A) Axial CECT image of the brain at initial presentation shows mild cerebral edema identified by narrow lateral ventricles and sulcal effacement (arrows). Serum sodium levels were less than 120 mEq/L. (B) Axial CECT image of the brain after slow correction of hyponatremia at a rate of 0.5 mmol of sodium per liter per hour shows resolution of the edema. (C) Axial CECT image of the chest shows a left lower lobe spiculated mass (arrow), proven by biopsy results to be SCLC. (D) Axial CECT image of the abdomen shows numerous hepatic metastatic lesions (arrowheads) and adrenal metastases bilaterally. Significant mesenteric lymphadenopathy (black arrow) and peritoneal carcinomatosis (white arrow) are also present.

some tumors, such as choline derivatives for prostate cancer and gallium-tagged DOTATATE for neuroendocrine tumors (55). Uncertain findings on plain radiographs on the underlying cause of HHM or unexplained abnormal radionuclide bone scan uptake should be evaluated with MRI (55).

Musculoskeletal PNSs

Tumors influence musculoskeletal tissues, including the periosteum, subcutaneous connective tissue, fascia, bones, and muscles. We describe herein musculoskeletal PNSs that have a temporal relationship to underlying tumors.

Hypertrophic Osteoarthropathy

Hypertrophic osteoarthropathy (HOA) is characterized by a chronic proliferation of the periosteum of long bones and digital clubbing due to ectopic production of growth hormone and vascular endothelial growth factor (7). Patients often present with arthralgia and symmetrical pain and swelling of the distal ends of their limbs. Primary HOA is

a rare genetic disease, while secondary HOA is much more common, comprising 95% of all cases (57).

The malignant tumors most commonly associated with HOA are pleural mesothelioma and non-SCLCs (58). Other nonpleural/pulmonary tumors include nasopharyngeal, esophageal, and breast carcinomas, as well as thoracic lymphoma. Benign acquired causes of HOA include aortic aneurysm, endocarditis, patent ductus arteriosus, pleural solitary fibrous tumor, and bronchiectasis (58,59).

Plain radiography should be the first imaging modality performed. Patients with HOA show periosteal reaction in the diaphysis of the tibia, fibula, radius, and ulna (Fig 13). In advanced cases, periostitis can extend to the metaphysis and epiphysis of all tubular bones (57). Localized periosteal reaction has only been described in benign causes of HOA. Typical soft-tissue radiographic findings of HOA include bulbous soft-tissue deformity at the distal fingers and abnormal nail curvature (57). Bone scintigraphy is the most sensitive test for HOA and will show increased paracortical

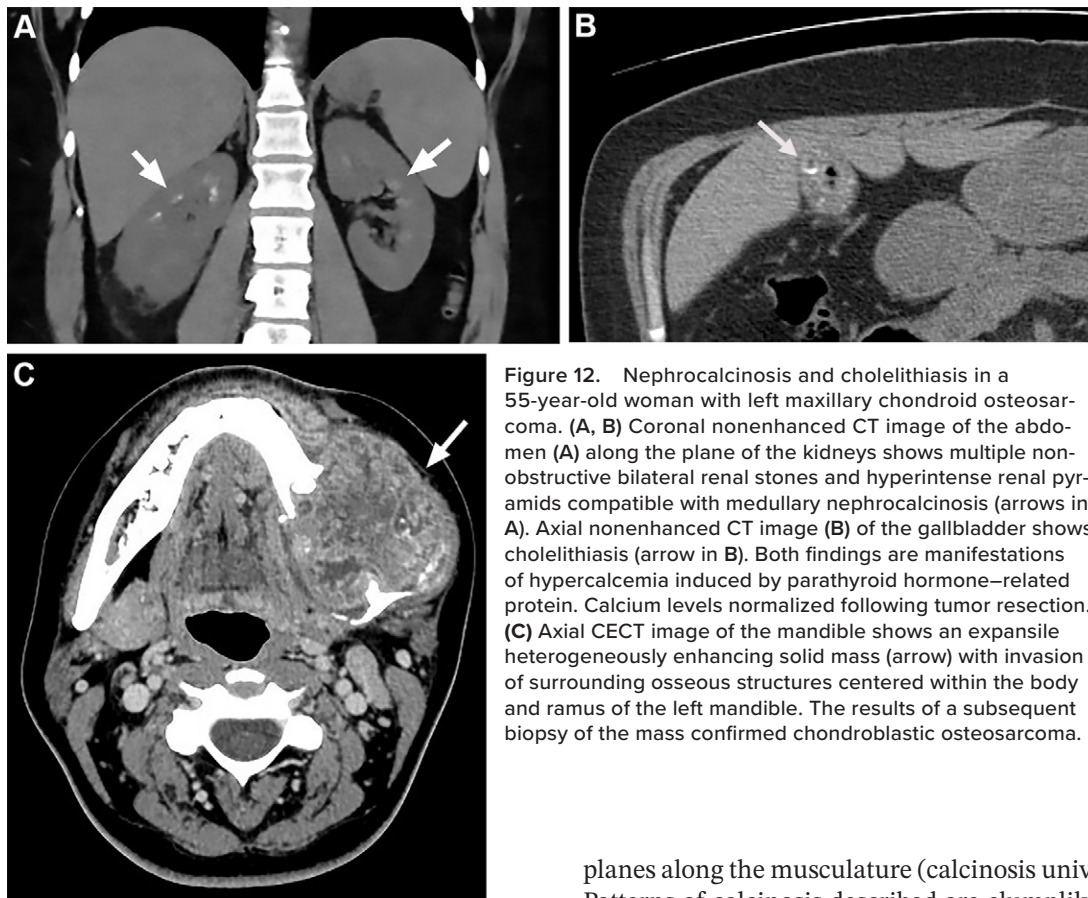


Figure 12. Nephrocalcinosis and cholelithiasis in a 55-year-old woman with left maxillary chondroid osteosarcoma. (A, B) Coronal nonenhanced CT image of the abdomen (A) along the plane of the kidneys shows multiple non-obstructive bilateral renal stones and hyperintense renal pyramids compatible with medullary nephrocalcinosis (arrows in A). Axial nonenhanced CT image (B) of the gallbladder shows cholelithiasis (arrow in B). Both findings are manifestations of hypercalcemia induced by parathyroid hormone–related protein. Calcium levels normalized following tumor resection. (C) Axial CECT image of the mandible shows an expansile heterogeneously enhancing solid mass (arrow) with invasion of surrounding osseous structures centered within the body and ramus of the left mandible. The results of a subsequent biopsy of the mass confirmed chondroblastic osteosarcoma.

radioactive tracer uptake with a symmetrical diffuse pattern in the affected bones (58). At MRI, periosteal reaction has low to intermediate signal intensity at T1-weighted imaging, with postcontrast enhancement and a low signal intensity at T2-weighted imaging. Furthermore, muscle swelling as well as septal and muscle edema can be demonstrated (57).

The differential diagnosis for HOA includes leukemia and lymphoma, where an associated marrow or soft-tissue abnormality can be appreciated at MRI, and a more prominent destruction is noted in Hodgkin lymphoma (57). Other differential diagnoses include venous stasis and antifungal treatment with voriconazole (57).

Dermatomyositis

Dermatomyositis is inflammatory proximal myopathy with multiple skin changes (60). Approximately 10%–25% of cases are paraneoplastic, with the commonly associated tumors being breast, lung, ovarian, and prostate cancers (1). For an acute presentation of dermatomyositis, early investigation of an underlying malignant neoplasm with use of chest and abdominopelvic CT or mammography and breast US is justified. However, extensive screening is not warranted in patients with long-standing dermatomyositis (7).

Radiographs of dermatomyositis show increased superficial skin radiopacity (calcinosis cutis) as well as increased radiopacity around joints that encounter frequent microtrauma, such as the elbows (calcinosis circumscripta) or deep fascial

planes along the musculature (calcinosis universalis) (Fig 14). Patterns of calcinosis described are clumplike, sheetlike, and reticular calcifications (61). In the chronic phase, characteristic calcifications appear in 25%–50% of cases, which may lead to localized muscle atrophy (62). MRI is the best modality for detecting soft-tissue abnormality (eg, active myositis and subcutis inflammation) with use of coronal and axial short τ inversion-recovery or T2-weighted sequences, especially with fat suppression. In contrast, CT is less accurate than MRI for the evaluation of muscle inflammation but is more sensitive than MRI in calcinosis detection (62).

Oncogenic Osteomalacia

Oncogenic osteomalacia is a rare PNS caused by bone and soft-tissue tumors. It is mainly caused by mesenchymal tumors, such as phosphaturic mesenchymal tumor, angiofibroma, and osteosarcoma (63). Osteomalacia is uncommon before the age of 40 years. Adults usually present with pathologic fractures and muscle weakness, while children present with gait abnormality, skeletal changes, and stunted growth (64). The inciting tumors are usually small and difficult to identify. Thus, FDG PET imaging, octreotide scintigraphy, and whole-body MRI have been used for tumor localization (65). Radiographs show signs of osteopenia (eg, Looser zones, diffuse osteopenia coarsening of trabeculae) and multiple fractures (3,65) (Fig 15).

The differential diagnosis of osteomalacia includes genetic causes (such as autosomal dominant/recessive hypophosphatemia, X-linked hypophosphatemia) that have a childhood onset or may be associated with a family history of these disorders. Other causes of osteomalacia include



Figure 13. HOA in a 66-year-old man with squamous cell lung cancer. (A) Lateral radiograph of the left ankle shows extensive periosteal reaction of the posterior cortex along the distal fibula (arrows). (B) Lateral radiograph of the left hand shows bone demineralization with subtle periosteal reaction of the cortex along multiple metacarpals and multiple proximal phalanges. Additional soft-tissue clubbing of the fingertips is present (arrows). (C) Axial CECT image of the chest shows multiple bilateral solid spiculated masses and nodules (arrows), of which the dominant mass was biopsy-proven squamous cell lung cancer.

Fanconi syndrome (urinary sugars and protein) and direct tubular renal damage by drugs or toxins (65).

Palmar Fasciitis and Polyarthrititis

Palmar fasciitis and polyarthrititis (PFPA) syndrome usually occurs in women and is characterized by a nonerosive polyarthrititis of the metacarpophalangeal and interphalangeal joints (66). Additional features consist of rapid inflammation and fibrosis of the palmar fascia leading to contracture of the hand (66). PFPA is usually associated with an underlying malignant tumor, such as that of the breast, fallopian tube, ovary, or endometrium. Prostate, stomach, and pancreatic cancers are other possibilities (3).

Imaging findings include narrowing of a joint compartment without erosions, as well as flexion deformities of the hand or fingers (Fig 16). Unlike Dupuytren disease, PFPA has a rapid progression and a higher severity of hand contracture (67). The noticeable induration of subcutaneous tissues seen in PFPA has resulted in the descriptive term

“woody hands” (67). Scleroderma may also mimic PFPA. However, patients with PFPA lack evidence of sclerodactyly, have rapid development of symptoms and hand disability, and less often have multiple additional affected joints (eg, knees and ankles) (67).

Dermatologic Manifestations of PNSs

PNSs can often be characterized by cutaneous lesions. Specific dermatologic findings, described in the following sections, can indicate the organs likely affected by malignancy, aiding the diagnostic search.

Necrolytic Migratory Erythema

Necrolytic migratory erythema consists of flaccid vesicles and bullae that progress to extensive erosions and crusting involving the groin, perianal area, and legs (25). It is strongly associated (in >70% of cases) with malignancies of the pancreatic islet cells (glucagonomas) (Fig E2) (68). Although necrolytic migratory erythema may occur with other

Figure 14. Dermatomyositis in three patients with underlying cancer who presented with muscle weakness and skin rash. (A, B) Axial nonenhanced CT image of the pelvis (A) at the level of the femoral heads in a 43-year-old woman with breast cancer who presented with skin rash and proximal muscle weakness shows extensive bilateral symmetric mineralization of the subcutaneous fat (arrows in A). Axial CECT image of the chest (B) shows a left upper outer quadrant spiculated solid breast mass with intralesional coarse calcifications (arrowhead in B) and associated marked skin thickening (arrow in B). (C) Axial fat-suppressed T2-weighted MR image of the pelvis in a 46-year-old woman with breast cancer shows symmetrically increased signal intensity in the subcutaneous fat over the right gluteus maximus muscle (arrows) compatible with edema, as well as low signal intensity in the right subcutaneous fat compatible with calcifications (arrowhead). (D) Lateral radiograph of the elbow joint in a 2-year-old boy with Wilms tumor shows a dense periarticular sheetlike pattern of calcifications indicative of dermatomyositis (arrows).

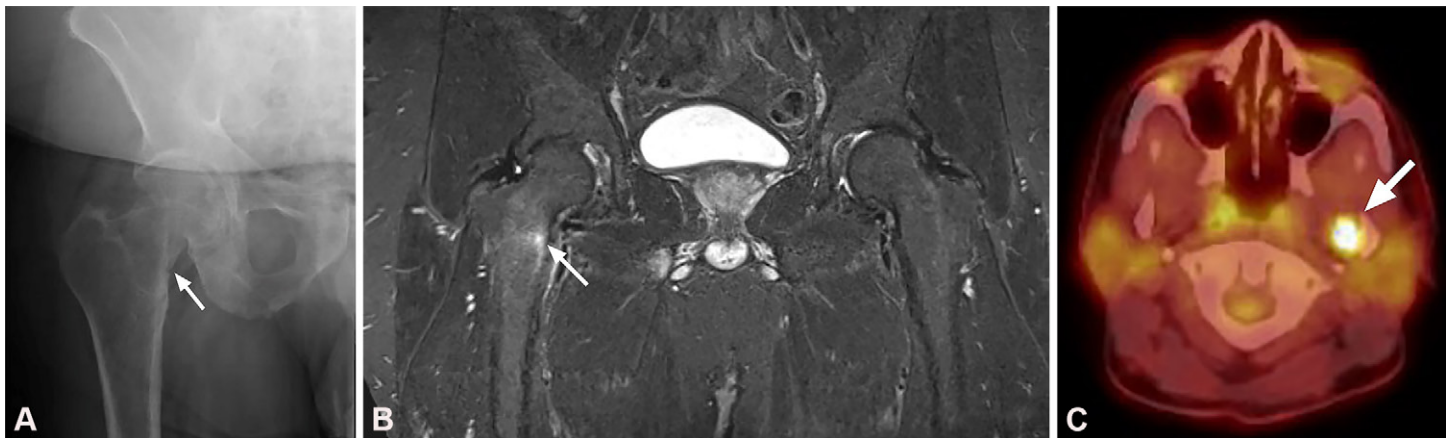
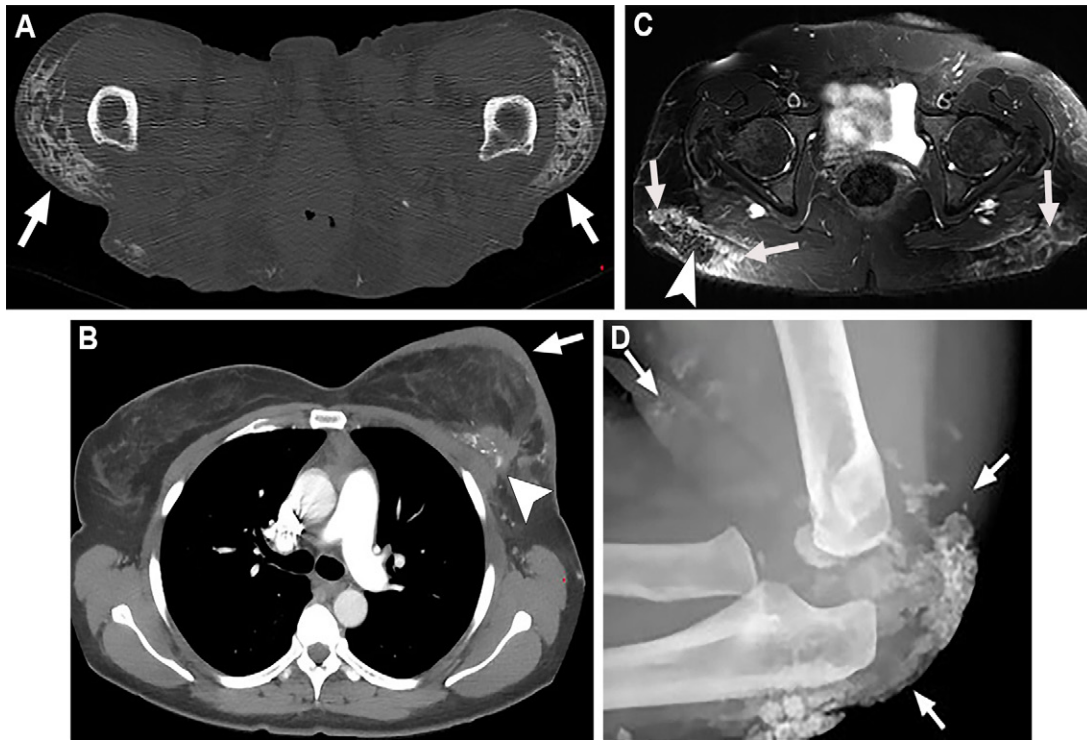


Figure 15. Osteomalacia and bilateral femoral stress fractures in a 50-year-old woman with persistent hypophosphatemia and elevated fibroblast growth factor 23 levels, supporting the diagnosis of a phosphaturic mesenchymal tumor. (A) Anteroposterior radiograph of the right hip shows small transverse cortical lucencies within the medial aspect of the right femoral neck (arrow), reflecting Looser zones, or pseudofractures, in the setting of osteomalacia. (B) Coronal fat-suppressed T2-weighted MR image of the hip shows corresponding high signal intensity along the medial aspect of the femoral neck, indicative of insufficiency fracture (arrow). (C) Axial color-fused gallium 68 DOTATATE PET/CT image shows a hypermetabolic left mandibular lesion (arrow) corresponding to a pathologically proven mandibular phosphaturic mesenchymal tumor.

nonmalignant conditions, the strong association with glucagonoma warrants initial investigation for pancreatic malignant neoplasms (7,68).

Ichthyosis

Ichthyosis is described as a scaly eruption with hyperkeratosis and is most commonly associated with Hodgkin



Figure 16. PFPA in a 49-year-old woman with right ovarian cancer. (A) Posteroanterior radiograph of the left hand shows flexion deformity involving the fourth and fifth proximal interphalangeal joints (arrows), which is indicative of palmar fascia contracture. (B) Oblique radiograph of the right hand shows severe erosive osteoarthritic changes involving the first carpometacarpal joint at the base of the thumb and the first, second, and third distal interphalangeal joints (arrows). (C) Axial T2-weighted MR image of the pelvis shows a predominantly solid heterogeneously hypointense pelvic mass (black arrow) with associated extensive peritoneal carcinomatosis (white arrows) and ascites (*).

lymphoma. Other associated malignant neoplasms include non-Hodgkin lymphoma, Kaposi sarcoma, leiomyosarcoma, and carcinomas of the ovary, breast, and cervix (25). Imaging-guided biopsy of lymph nodes is usually needed and helps to establish the diagnosis of lymphoma by confirming atypical lymphoid cells. PET/CT is also justified to determine the distribution of lymphadenopathy and assess the patient's response to treatment (Fig E3) (69).

Porphyria Cutanea Tarda

Porphyria cutanea tarda is characterized by bullae and vesicles that form in response to minor trauma or light sensitivity. Baravelli et al (70) showed that patients with porphyria cutanea tarda have a 20-fold increased risk of hepatocellular carcinoma and a sevenfold increased risk of biliary tract and gallbladder cancer. Patients with porphyria cutanea tarda should undergo screening with abdominal US and serum α -fetoprotein testing. A suspected hepatic lesion can be further evaluated with a dedicated CECT or MRI examination (71).

Role of PET/CT in PNSs

Patients with suspected PNSs may have an underlying early-stage cancer that is difficult to detect at conventional imaging (72). PET/CT has shown high sensitivity and negative predictive value and may serve as a whole-body screening tool for the underlying tumors, particularly in patients with negative conventional imaging findings (73). Pena Pardo et al (6) suggested a clinical pretest risk stratification system to select candidates with PNSs to undergo PET. Meanwhile, Vaidyanathan et al (74) concluded that a low clinical score does not exclude an underlying tumor, and PET findings can be concordant with, upgrade, or downgrade the clinical risk scoring. Nevertheless, the proportion of new malignant neoplasms identified at PET increases as the clinical risk of PNSs increases (6,74).

Although a few studies showed that PET could be used to identify tumors in PNS settings that were missed at conventional imaging (74,75), not enough data are present to evaluate the additional screening benefit of PET over conventional modalities and investigate its cost-effective value (72,73). Similarly, more research is needed to follow up on patients with negative

PET screening results, given its high negative predictive value and the paucity of available PET follow-up studies in this patient population (72,74).

Conclusion

PNSs are a heterogeneous set of conditions that indicate the presence or precede the development of a tumor. They are relatively rare disorders that require exclusion of more common conditions. Knowledge of the spectrum of the underlying tumors of PNSs substantially helps tailor diagnostic imaging for patients. Radiologists should be aware of imaging findings and potential diagnostic pitfalls of PNSs, as this can enable timely diagnosis of underlying tumors and positively impact clinical outcomes for these patients.

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